

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0806

Parleys 138 kV
September 28, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0806. The Interconnection Customer proposes a 149.4 MW solar photovoltaic facility interconnecting at the Parleys 138 kV substation in the Pennsylvania area. The request was received on June 09, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 330 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$49,123,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Chase Hill to Reclaim 138 KV	Burwell to P.T.S Substation 345 KV	227.0	210	5.5
Reclaim to Centerville Rd 138 KV line	P.T.S to Burwell 345 KV	213.0	233	5.5
Centerville Rd to San Carlos 138 KV	P.T.S to Burwell 345 KV	208.0	244	5.5
Ontonagon Substation to Roseland 138 KV	Hilltop to Piney Creek Project 500 KV	207.8	179	8.0
Somerville to San Carlos 138 KV	Burwell to P.T.S 345 KV line	204.7	252	5.5
Wilmerding to Dravosburg 138 KV	Medora to P.T.S 345 KV line	185.4	252	37.9
Roseland to Mon Vly Works Illinois 138 KV line	Hilltop to Piney Creek Project 500 KV	184.2	198	8.0
Gregg to Dravosburg 138 KV	Medora to P.T.S 345 KV	173.3	279	37.9
Parleys to Gregg 138 KV	Medora to P.T.S 345 KV	170.3	288	37.9
Wilson 1&2 345 kV/138 kV Autotransformer	Burwell 500/345 kV Transformer Bank	164.9	374	6.8
Dos Amigos to Ontonagon 138 KV L	Lewisville to Williams 138 KV	164.3	239	15.4
Thermas Substation to Wilson 1&2 138 KV	Burwell 500/345 kV TX	160.7	323	5.9
Mon Valley Works Illinois to V C Summer 138 KV	Hilltop to Piney Creek Project 500 KV line	160.5	218	8.0
Wilmerding to Drummond 138 KV	Burwell to P.T.S 345 KV Line	156.9	244	15.2
Knoll Substation to Drummond 138 KV	Burwell to P.T.S 345 KV	145.6	254	15.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Wineagle Substation to Augusta Electric Plant No 1 Substation 138 KV	Matanzas to Payne 138 KV	141.9	242	5.0
Burwell to Knoll 138 KV	P.T.S to Burwell 345 KV	135.2	261	15.2
Spring Ridg Rd to V C Summer 138 KV	Piney Creek Project to Hilltop 500 KV	131.7	246	8.0
Spring Ridge Rd to E Omak Substation 138 KV Line	Piney Crk Project to Hilltop 500 KV	127.2	250	8.0
Wilmerding to Delco Substation 138 KV line	Medora to P.T.S 345 KV L	117.2	221	24.6
E Omak to Station Q Harbor 138 KV	Piney Creek Project to Hilltop 500 KV Line	116.0	260	8.0
Yukon to Sta Q Harbor 138 KV line	Piney Crk Project to Hilltop 500 KV	111.8	262	8.0
Youngwood 500 kV/345 kV Auto	Univ Of Nc Chapel Hill Cogen Facility to Youngwood 345 KV	109.9	732	5.0
Yukon to Univ Of Nc Chapel Hill Cogen Facility 138 KV	Hilltop to Piney Crk Project 500 KV	108.2	265	8.0
Kearny to Seward 345 KV	Fort Gibson to Armstrong 500 KV	107.5	1,386	7.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Reclaim to Chase Hill 138 KV	Rebuild line with higher-capacity conductor	\$1,664,000	30% (\$491,000)
Reclaim to Centerville Rd 138 KV line	Rebuild line with higher-capacity conductor	\$2,655,000	30% (\$784,000)
San Carlos to Centerville Rd 138 KV	Rebuild line with higher-capacity conductor	\$842,000	30% (\$249,000)
Rosland to Ontonagon 138 KV L	Rebuild line with higher-capacity conductor	\$1,590,000	100% (\$1,590,000)
Somerville to San Carlos 138 KV	Rebuild line with higher-capacity conductor	\$1,552,000	30% (\$458,000)
Wilmerding to Dravosburg 138 KV	Rebuild line with higher-capacity conductor	\$6,259,000	78% (\$4,889,000)
Roseland to Mon Vly Works Illinois 138 KV line	Rebuild line with higher-capacity conductor	\$3,638,000	100% (\$3,638,000)
Gregg to Dravosburg 138 KV	Rebuild line with higher-capacity conductor	\$3,265,000	78% (\$2,550,000)
Parleys to Gregg 138 KV	Rebuild line with higher-capacity conductor	\$704,000	78% (\$550,000)
Wilson 1&2 345 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$13,861,000	29% (\$4,030,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dos Amigos to Ontonagon 138 KV L	Rebuild line with higher-capacity conductor	\$1,408,000	100% (\$1,408,000)
Thermas to Wilson 1&2 138 KV Line	Rebuild line with higher-capacity conductor	\$3,667,000	32% (\$1,170,000)
Mon Valley Works Illinois to V C Summer 138 KV	Rebuild line with higher-capacity conductor	\$2,751,000	100% (\$2,751,000)
Wilmerding to Drummond 138 KV line	Rebuild line with higher-capacity conductor	\$2,210,000	100% (\$2,210,000)
Knoll Substation to Drummond Substation 138 KV	Rebuild line with higher-capacity conductor	\$2,299,000	100% (\$2,299,000)
Wineagle Substation to Augusta Electric Plant No 1 Substation 138 KV	Rebuild line with higher-capacity conductor	\$3,138,000	17% (\$534,000)
Burwell to Knoll 138 KV	Rebuild line with higher-capacity conductor	\$1,318,000	100% (\$1,318,000)
Spring Ridge Road to V C Summer 138 KV Line	Rebuild line with higher-capacity conductor	\$2,563,000	100% (\$2,563,000)
Spring Ridge Rd to E Omak Substation 138 KV Line	Rebuild line with higher-capacity conductor	\$2,237,000	100% (\$2,237,000)
Wilmerding to Delco Substation 138 KV line	Rebuild line with higher-capacity conductor	\$1,323,000	100% (\$1,323,000)
Sta Q Habor to E Omak 138 KV	Rebuild line with higher-capacity conductor	\$4,842,000	100% (\$4,842,000)
Yukon to Sta Q Harbor 138 KV line	Rebuild line with higher-capacity conductor	\$2,940,000	100% (\$2,940,000)
Youngwood 500 kV/345 kV Transformer	Replace transformer with higher-rated unit	\$35,990,000	9% (\$3,373,000)
Yukon to Univ Of Nc Chapel Hill Cogen Facility 138 KV	Reconductor line	\$843,000	100% (\$843,000)
Kearny to Seward 345 KV	Reconductor line	\$2,823,000	3% (\$83,000)
	Total allocated to project		\$49,123,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.